Assessing Quantum Machine Learning on Small Datasets: Accuracy, Execution Time, and Practical Usability

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Abstract—**This work aimed to analyse quantum machine learning algorithms available in the IBM Qiskit QML framework [1], comparing them with classical models running on occasional machines, evaluating the effects of noise and the ability to provide solutions, as well as exploring variational quantum circuits [2] that operate in a hybrid way, together with a quantum platform and a. One of the biggest obstacles to the advancement of quantum computing is noise, which generates errors and significantly compromises the results [3]. Variational quantum circuits seek to mitigate this problem, offering greater accuracy in the results [2]. Pure quantum models and quantum models with variational circuits were tested, in addition to classical models, using the Iris dataset [4]. Tests of quantum algorithms were performed on IBM quantum computers with 156 fixed-frequency qubits and electrical couplings, from the Heron conversion family, currently the highest performing, and the core of the System Two Architecture [5], [6], using the AerSimulator simulation library [7] running locally on PC I7 architecture in an MS Windows environment. The same machine was used to run the classical algorithms. The results obtained demonstrate that there are still many challenges to be overcome for the declarations of quantum computing architectures [55], although the hardware roadmap has been advancing considerably in recent years [8].**

Keywords— Quantum Machine Learning, Iris Dataset, Quantum Theory.

I. Introduction

In 2019, Google announced quantum supremacy with the Sycamore quantum processor, capable of performing calculations in minutes that would take 10,000 years for a classical supercomputer of the time to execute [10]. For many years we have heard about quantum computers and their achievements, but what practical applications do these computers have in today's world? Could we use them in daily life, or even for more complex practical problems? After all these years, where are we? The roadmap of many large companies in the sector has shown positive results, yet there are few announcements or companies applying this technology in everyday commercial or scientific use that are visible to the public.

Some of obstacles help explain this gap. First, the development of quantum algorithms did not advance as much as hardware development; quantum algorithms are difficult to implement due to the latent problem of noise that generates decoherence in qubits [7]. Second, quantum processors are generally susceptible to errors generated by internal noise in the chips. For instance, even when cooled to extremes, task execution generates a small amount of residual heat that interferes with the quantum entanglement of the gates, resulting in inaccuracies [12]. In the current era of noisy intermediate-scale quantum (NISQ) devices, the coherence time of qubits is limited to milliseconds, insufficient time for complex calculations, since coherence and entanglement are fundamental to quantum computation [13], [12].

This work analyses current quantum algorithms that can be used in machine learning, assessing the performance of quantum execution against the computation of the same algorithms on classical computers, while examining accuracy, cost, and the possibility of applying quantum computing in companies. IBM quantum computers are used, and the models run on IBM's Qiskit library [1], [14], so that we can determine under which conditions quantum models can offer practical advantages over consolidated classical models on a small-scale dataset The IBM quantum computers used have up to 156 qubits; however, only 2 qubits were used for all tests. Although this may seem like a misguided decision, it had strong practical and strategic reasons, especially due to the Iris dataset. The decision rests on the balanced use of resources. The time constraints of 10 minutes to run all applications on a real quantum processor demanded this, and due to the number of variables in the Iris dataset (4 features, 3 classes, 150 samples), models with 2 qubits are more than sufficient to achieve high precision, as we are dealing with dependent and independent variables. Adding more qubits to simple machine learning models would only multiply the number of qubits affected by noise [55]. Quantum computers are very noisy, making noise one of the biggest obstacles to the use of QML, since increasing the number of logic gates, in this case deeper circuits, leads to an accumulation of noise throughout the problem solution. According to Samson Wang [?], the complexity of the problem should define the number of qubits to be used. This leads to the research question that motivates the study:

"Under which conditions can quantum machine learning models offer practical advantages in terms of accuracy, execution time, and usability, over consolidated classical models on small datasets in the NISQ era?"

The null hypothesis (H_0) assumes no appreciable difference between the classical and quantum paradigms, while the alternative hypothesis (H_1) states that a measurable difference, attributable to the paradigm and the execution conditions, does exist. Classical models are used as a control so that we can analyse the accuracy gap that could justify the use of quantum computers and low-dimensionality models, and discuss the limitations imposed by noise in the current NISQ era [12].

II. Background and Related Work

A. Historical Perspectives

The accessibility of quantum computing today is the latest step in a long lineage of computing democratisation. Seymour Cray's supercomputers advanced numerical and data processing through innovations such as freon cooling, integrated circuits and gallium-arsenide semiconductors, at enormous research cost [15]. The subsequent "WINTEL" era moved the market towards parallel commodity processors, and Big Data later turned clusters of ordinary machines into supercomputers, popularising machine learning in industry. The exponential growth of storage, RAM and SSDs, together with cloud architectures, allowed even small companies to embed predictive algorithms into their core business; quantum cloud platforms now extend that same democratising trajectory to quantum hardware.

The conceptual roots of the field were traced to Richard Feynman, who in his 1981 paper argued that nature is itself a great quantum computer and that simulating physics would demand machines able to compute massively in parallel [9], a vision that motivated architecture such as the Connection Machine [16]. In 1994 Peter Shor and Don Coppersmith introduced the quantum Fourier transform underlying Shor's factoring algorithm [11]. Quantum machine learning itself was recently consolidated: Biamonte et al. [17] and Schuld et al. [18], [19] synthesised the field, distinguishing methods that accelerate the linear algebra underlying learning from those that train parametrised circuits, the latter being the focus of this work.

B. Quantum Machine Learning for Classification Tasks

Considering previous works related to quantum machine learning, we can start with the first approach, which is the attempt to accelerate the mathematical calculations used in classical algorithms, including in this context the more efficient use of quantum memories (QRAM) for this type of activity. According to [17],[19], the field can be organised into different approaches. One of these is the training of quantum circuits with adjustable parameters, which is the approach adopted in this work. In [20], [21], the authors proposed increasing exponential speed gains for the quantum version of SVM by considering the use of QRAM. In [22], the author stated that even these gains in a quantum processing environment could be achieved by classical algorithms based on data access conditions in an efficient way. Based on these contexts above and the availability presented by the IBM Quantica architecture, the work opted for the second approach described below, which focuses on the use of trainable circuits without dependence on Quantica memories.

The second approach, based on the work in [23] carried out a classification method based on "quantum feature spaces" that obtained gains with the between classes, which was tested only on small and artificial datasets, using old quantum hardware. In [24], the authors compared variational quantum classifiers for binary and multiclass classification. Their results showed that the binary classifier achieved higher accuracy than the multiclass versions. These findings motivated the use of binary classification in this work, which is implemented using a two-qubit quantum circuit. The work of [25] comprehensively reviewed the use of variational quantum algorithms and encountered what are called sterile plateaus, variational quantum algorithms rely on machine learning in classical computers that are responsible for generating values that are selected in quantum computers from the results to define a path to be followed by the quantum machine. The values generated in quantum computers sometimes fail to define a correct path to follow in the calculation process due to noise. This creates situations where the result does not reach the minimum precision required in the calculation. To avoid this type of problem, limiting the number of variables to the number of qubits can give the result greater precision, since an excess of qubits accumulates errors generated by noise, directly impacting the calculation. This was the main reason for the approach to using two variables or two qubits in the work, that will be later presented in section IV.

C. Hardware Platforms Used in QML Literature

The choice of hardware for this QML study had a direct influence on the results. The superconducting gate-based quantum processors provided by IBM and Google, which are universal quantum computers, are the architectures most widely adopted in the classification literature reviewed [17]–[19], [23], [24], [26]. Furthermore, these platforms are among the most accessible, as they provide free cloud access. However, it is important to note that there are limitations on usage time and available resources.

Other quantum hardware architectures have also demonstrated promising characteristics. Google's Willow processor was mentioned as one of the most capable of suppressing errors exponentially below the limits necessary for error correction, while Quantinuum based on trapped ions were reported as having greater logic qubits with greater fidelity than the physical qubits of which they are composed. Consequently, trapped ion-based systems offer greater coherence time, consequently greater fidelity in logic gates, being cited in several works [13]. In contrast, the D-Wave hardware architecture was used by [27] for combinatorial and logistic optimisation problems, but not in classification, as the quantum annealing architecture does not allow feature maps and ansätze. In addition, Photonic architectures were not found in the literature selection for use in QML classification to date, making any type of comparative study impossible.

D. Practical Applications

Quantum machine learning was consolidated as a formalised discipline only recently. In [17], the authors distinguish approaches that accelerate the linear algebra underlying learning from those that use trainable parametrised circuits, and part of these implementations, available within Qiskit's QML libraries, present theoretical gains by relying on efficient data access (QRAM) that are not always feasible in practice [20]–[22]. On small, low-dimensional datasets the most common applications are supervised classification with variational quantum classifiers and kernel methods that embed data in Hilbert space [1], [12], together with unsupervised clustering through quantum analogues of k-means such as q-means [28], [29].

These techniques have been explored across several fields. One area gaining attention is the simulation of materials and molecules: because such elements have multidimensional structures, classical software complexity grows sharply, whereas quantum computers operate naturally within the multidimensional Hilbert space, opening opportunities for simulating new drugs and cellular behaviour. D-Wave, using an annealing architecture, has pursued machine-learning optimisation for logistics and transport [27]. More recently, hybrid quantum–classical pipelines have been applied to the Iris dataset itself, including variational classifiers [24] and hybrid quantum convolutional networks [26], which makes it a suitable and widely used benchmark for paradigm comparison.

A frequently cited opportunity is the simulation of molecules and materials. Because such systems have multidimensional structure, classical simulation scales poorly, whereas quantum computers operate within the multidimensional Hilbert space; the drug-metabolism enzyme cytochrome P450, for instance, has a structure that maps naturally onto a quantum model [23]. This is precisely the regime in which a genuine quantum advantage is expected, unlike the small, low-dimensional classification task studied here, which is included specifically to probe the lower bound of usefulness on commodity problems.

Recent milestones illustrate the pace of progress. In 2024, Google's Willow chip with 105 qubits demonstrated, for the first time, an exponential suppression of errors below the surface-code threshold, while Quantinuum produced multiple logical qubits with fidelity higher than the underlying physical qubits. In communication, China launched the Micius satellite in 2016 and, by 2020, an integrated space–ground network; the China Quantum Communication Network now spans more than 10,000 km with 145 backbone nodes [30]. These achievements show that the hardware and networking layers are maturing, even as a comparable breakthrough on the algorithmic side, capable of delivering an everyday commercial advantage, remains elusive.

The 2019 supremacy claim was promptly contested: IBM argued that its Summit supercomputer could reproduce the same calculation in about 2.5 days rather than 10,000 years, so that the 200-second quantum runtime represented only a temporary advantage on a circuit with no practical application [31]. In communication, China launched the Micius satellite in 2016 and demonstrated quantum key distribution in 2017 and is today at the forefront of quantum networking [32]. These results reinforce that the hardware frontier is advancing rapidly even where a practical algorithmic payoff is still pending.

E. *Constraints of Current Quantum Hardware*

Despite an optimistic roadmap, limitations still exist in terms of adoption in the context of everyday commercial and scientific use of quantum computers. In 2025, HSBC announced a 34% improvement in the probability of winning clients in the European corporate-bond market in partnership with IBM [33], and the Brazilian bank Itaú announced that it is exploring quantum computing in its financial operations [34]; yet at present there is no major discovery, much less large-scale deployment, whose results would justify it. The hardware has advanced greatly, but a large gap remains on the algorithm side. Aside from Shor's [11] and Grover's algorithms, few are able to exploit quantum resources effectively.

For small datasets the constraints are even more acute. The central problem is the absence of a noise-free universal quantum computer: noise is directly related to the coherence time of qubits, and short entangled states prevent more complex results from being achieved. The lack of QRAM for storing quantum results further restricts longer applications, a direct consequence of NISQ limitations [35], [12]. To run on NISQ hardware, the problem must be reduced to very few qubits, which discards information and brings classes closer together; combined with a small number of measurement shots and the eight-iteration ceiling imposed on variational training, this caps the accuracy attainable on a small problem such as Iris. Network latency between the local machine and the remote QPU, together with queueing on shared cloud resources, adds a further practical penalty that dominates the wall-clock time.

The challenges are well understood. On the hardware side, better qubits with lower error rates are needed, since one cause of noise is the proximity of qubits, which leak signal and heat into the substrate and induce decoherence; more qubits, improved cooling and supercooling, and error correction, grouping several physical qubits into one logical qubit, or using variational circuits in a hybrid scheme, are equally critical. On the software side, quantum algorithms are still young and demand intensive optimisation, hybrid quantum–classical integration must be made efficient, and better error-mitigation software could reduce the dependence on dedicated correction hardware. For a small problem such as Iris, these constraints translate directly into the two-qubit limit, the small number of shots, and the short variational training used in this study.

Beyond hardware and software, adoption depends on people. More scientists and researchers are needed, alongside greater access to the machines, and quantum-physics concepts must reach schools and universities so that future developers can use these computers effectively. Developers face the additional hurdle that not all quantum computers are compatible, although transpilers and adaptable libraries increasingly ease migration; ultimately, wider adoption requires lower usage cost, greater availability and a gentler learning curve.

F. *Democratisation and Software*

Having covered the use cases and the hardware context, the question of usability concerns the second part of the title, accuracy, execution time, and practical usability, for every type of user. Today all major cloud providers offer access to quantum computers with a minimum free allocation for testing, alongside compiler technologies documented by those providers. Languages and libraries such as Qiskit, Cirq and PennyLane have matured year on year, complemented by simulators that run locally on the developer's machine or in a distributed manner on cloud instances using graphics accelerators for significant simulation performance.

This democratisation lowers the barrier to entry, but usability is not yet trivial. The technology demands a more robust education in quantum-physics concepts, and although transpilers and adaptable libraries increasingly allow systems to migrate between different quantum computers, not all machines are compatible. For the business analyst and end user, the final element in the adoption chain, what ultimately matters are whether the accuracy and execution time delivered on a real device justify the cost. The remainder of this paper measures exactly that, using a reproducible benchmark in which classical models act as a control against seven quantum and hybrid models.

Usability, in practice, is inseparable from price: at a list cost on the order of US$100 per minute of Quantum Processing Unit (QPU) time (year of 2026), the execution-time overhead measured later in this paper is not merely a performance footnote but the decisive barrier to adoption for small-scale problems.

From a tooling standpoint, the same Qiskit code path runs first on the local AerSimulator and then, unchanged, on the remote QPU, with the backend, credentials, shot count and hardware guards all declared in a typed configuration file. This separation of concerns is what makes the experiment reproducible and what allows a single researcher, on commodity hardware, to exercise an industrial superconducting processor, the practical face of democratisation, even when the measured performance does not yet justify the cost.

Based on all the studies researched [17], [18], [19], [23], [24], [26] the great contribution of the current work was to have tested classical, quantum and variational models together in a single work and application, being able to have all the accuracy results of these models in a comparative way, something that was presented in a discreet way in the other works.

III. Methodology

A. Computational Elements and Environment

Based on the findings from the literature review, IBM's superconducting gate-based quantum architecture was selected for this study. Specifically, the experiments were executed using the Heron r2 quantum processor, a universal gate-based quantum computer that is representative of the hardware adopted in the majority of previous QML classification studies [23], [24], [26]. All quantum algorithms were intensively tested in simulation mode using AerSimulator, which is the framework that allows simulating quantum processors running on Qiskit. Unlike previous studies, this work provides a comprehensive comparison of classical machine learning, quantum machine learning, and variational QML algorithms using the same Qiskit dataset and experimental pipeline. The complete implementation, including the source code and experimental results, is publicly available on GitHub [37].

All classical and quantum models were executed within a single experimental framework that generated structured execution logs and a consolidated results file (comparison_results.json). The experiments were performed on a Dell notebook with an 11th-generation Intel Core i7-1185G7 processor running at 3.00 GHz, 32 GB of RAM and 1 TB SSD. The quantum component ran on the ibm_marrakesh QPU, which was selected automatically through Qiskit Runtime based on queue availability.

The quantum architecture used was the IBM platform based on the Heron r2 QPU; the QPUs available carried 156 qubits and were hosted at the IBM data centre in Washington, USA. IBM quantum computers use superconductor-based QPUs, selected in real time through the QiskitRuntimeService package and a typed configuration file (.env) that exposes the backend name, credentials, number of shots, transpilation optimisation level, variational hyperparameters, and hardware guards.

The software stack combined Qiskit (1.3–1.x) for circuit construction and transpilation; qiskit-machine-learning (0.8.x) for VQC, QSVC, SamplerQNN and fidelity kernels; qiskit-ibm-runtime (0.30+) for submitting jobs to the QPU through SamplerV2; qiskit-aer (0.14+) for local simulation and device noise models; scikit-learn (1.3+) for the classical models, metrics, PCA and scalers; NumPy/pandas for numerical and tabular manipulation; matplotlib/seaborn for visualisations; and python-dotenv for loading the typed configuration.

IBM's QML library exposes a programming model close to that of scikit-learn or PyTorch, which lowers the entry barrier. A very important feature for adapting new generations of programmers to the world of quantum computing. The ease of use of QML greatly favoured this work, enabling the development of the entire application used in the tests.

B. Dataset and Pre-processing

The chosen dataset for the assessment of the machine learning models in this work was the Iris dataset [4]. It was popularised by the British biologist and statistician Ronald Fisher in his 1936 article on the use of multiple measurements in taxonomic problems, a classic example of linear discriminant analysis [4]. It contains 150 samples distributed across three species, Setosa, Versicolor and Virginica, organised by morphological attributes (sepal and petal length and width). Pre-processing followed four deterministic steps: standardisation, dimensionality reduction, angle encoding, and a stratified partition, each described below.

The dataset was originally compiled by Edgar Anderson to quantify the morphological variation of iris flowers across three related species, two of which were collected on the Gaspé Peninsula, and it has since become a standard reference in teaching and in machine-learning testing [36][4]. Its modest size and clear class structure make it well suited to a controlled comparison in which the same pre-processing pipeline is applied identically to every classical and quantum model.

C. Standardization

The four attributes are standardised with scikit-learn's StandardScaler, a Z-score normalisation that removes the mean and scales to unit variance, $x' = (x - \mu) / \sigma$. This standardisation is necessary for the subsequent Principal Component Analysis (PCA) and angle-encoding steps, which are sensitive to attribute scaling. PCA then reduces the four dimensions to two components: the goal was to reduce the number of qubits (NUM_QUBITS, default 2), lowering circuit dimensionality and noise exposure to enable execution on NISQ hardware. This reduction discards information and brings the Versicolor and Virginica classes closer together, a deliberate trade-off imposed by the limited QPU time available.

The two principal components are then rescaled to the interval $[0, \pi]$ with MinMaxScaler to prepare angle encoding. Working with angles rather than raw numerical values avoids periodicity ambiguities between qubit vectors, as represented on the Bloch sphere. Finally, an 80%/20% stratified split (TRAIN_SPLIT) preserves the three classes in all partitions, and a fixed random seed (RANDOM_SEED, default 42) guarantees exact reproducibility. Table I presents all the models tested in the experiment, including classical, quantum (non-variational), and quantum variational models.

On the Bloch sphere, a single un-entangled qubit state without global phase is written $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi} \sin(\theta/2)|1\rangle$, with rotations about the x, y and z axes implemented by the corresponding rotation-operator gates [33]. Encoding the two principal components as rotation angles places each sample at a well-defined point on the

TABLE II. ANALYSED MODELS

Model	*Paradigm*
LogReg	Classic
SVM	Classic
kNN	Classic
MLP	Classic
QkNN Quantum	non-variational
VQC Quantum	variational
QSVM Quantum	non-variational
QCL Quantum	variational
QFC Quantum	non-variational
QNN Quantum	variational
QFE-LR Quantum	non-variational

sphere, which is the geometric basis of the angle-encoding step.

D. Experimental Setup

Quantum computers are probabilistic machines: each execution of a circuit produces a superposition of outcomes, but every measurement collapses the wave function and reads a single result, so the circuit must be repeated. Running 1024 shots therefore executes the circuit 1024 times and accumulates the readings, some of which are discrepant because of noise from qubit heating, qubit proximity, or external electromagnetic interference. IBM's platform offers up to 156 qubits with roughly ten minutes of free QPU time per month, sufficient for testing once the application has been debugged on the local AerSimulator [34].

Hardware guards, safety limits declared mostly in the .env file, are activated automatically when an execution is detected to target a real QPU rather than the simulator, controlling cost and time. The most consequential guard limits the COBYLA optimiser to eight iterations. The quantum component was executed on the ibm_marrakesh superconducting QPU; of its 156 physical qubits only two were used, with a small, stratified test subset of six samples (two per class) imposed by the guards.

Because measurement on a quantum device is destructive, accuracy depends on the number of shots: each of the 1024 rounds executes the circuit and reads a single outcome, and only by accumulating many rounds does the empirical distribution emerge from the noise. A classical model obtains its answer in a single pass, whereas the quantum models must repeat the circuit many times, one of the structural reasons for the large cost gap reported in the results.

E. Models Compared

The benchmark compares four well-established classical models against seven quantum and hybrid models, all executed within the same run. The classical baselines use scikit-learn with fixed hyperparameters and serve as a control (see Table II). The quantum models are divided into three functional groups: variational classifiers, kernel/fidelity methods, and non-variational baselines (see Table III).

Variational models combine three configurable parts: feature maps (ZZFeatureMap, PauliFeatureMap [1], [12]) for non-linear encoding in Hilbert space; ansätze such as RealAmplitudes, EfficientSU2 and TwoLocal [38] with full entanglement; and classical optimisers (COBYLA [13], SPSA [23], L-BFGS-B, ADAM) that adjust the ansatz parameters to minimise the cost function. QCL and VQC pair a ZZFeatureMap with RealAmplitudes optimised by COBYLA, while QNN is built on a SamplerQNN with an EfficientSU2 ansatz and a parity interpretation that maps measured bit strings to class labels; each has cost O(MAX_ITER × n).

TABLE I. CLASSICAL REFERENCE MODELS AND CONFIGURATION.

Model	*Type*	*Principal hyperparameters*
SVM (RBF)	Maximum margin	kernel=RBF, C=1.0, gamma=scale
k-NN	Neighborhood	k=5, uniform weights, Euclidean
Logistic Reg.	Linear	solver=lbfgs, C=1.0, max_iter=MAX_ITER
MLP	Neural network	layers (64,32), ReLU, early stopping

TABLE III. QUANTUM MODELS, CLASSICAL ANALOGUES, AND VARIATIONAL CLASSIFICATION.

Model	*Classic analogue*	*Var.?*	*Characterization*
QSVM	SVM	No	Quantum fidelity kernel + SVC
QkNN	k-NN	No	Similarity by fidelity (or proxy)
QCL	Logistic Reg.	Yes	ZZFeatureMap+RealAmplitudes, COBYLA
VQC	MLP	Yes	Variational classifier
QNN	-	Yes	SamplerQNN (EfficientSU2 + parity)
QFC	-	No	Average-fidelity classifier (baseline)
QFE-LR	-	No	Fixed embedding + logistic regression

QSVM is a support-vector machine whose kernel is the quantum fidelity between encoded states, computed by a ComputeUncompute circuit [1], [12], and QkNN measures similarity by fidelity (or a Gaussian proxy). The two non-variational baselines, QFC (classification by average fidelity) and QFE-LR (a fixed quantum embedding followed by logistic regression), are inexpensive references against which the trained variational circuits can be compared.

To understand a variational quantum circuit, it helps to recall the notion of a Hamiltonian, the sum of the energies of a system. A variational circuit is defined by an ansatz (German for "initial attempt"), a parametrised circuit whose parameters are tuned to minimise a cost function associated with that Hamiltonian. The circuit operates in a hybrid loop: a classical optimiser proposes parameter values, the quantum device evaluates the circuit and measures an expectation value, and the result is returned to the classical side, which refines its guess until convergence. More shots and iterations generally improve precision, but they also raise cost. Because training runs on the local classical machine, the latency between Ireland and Washington and the limited numerical power of the notebook penalise these models heavily a high-

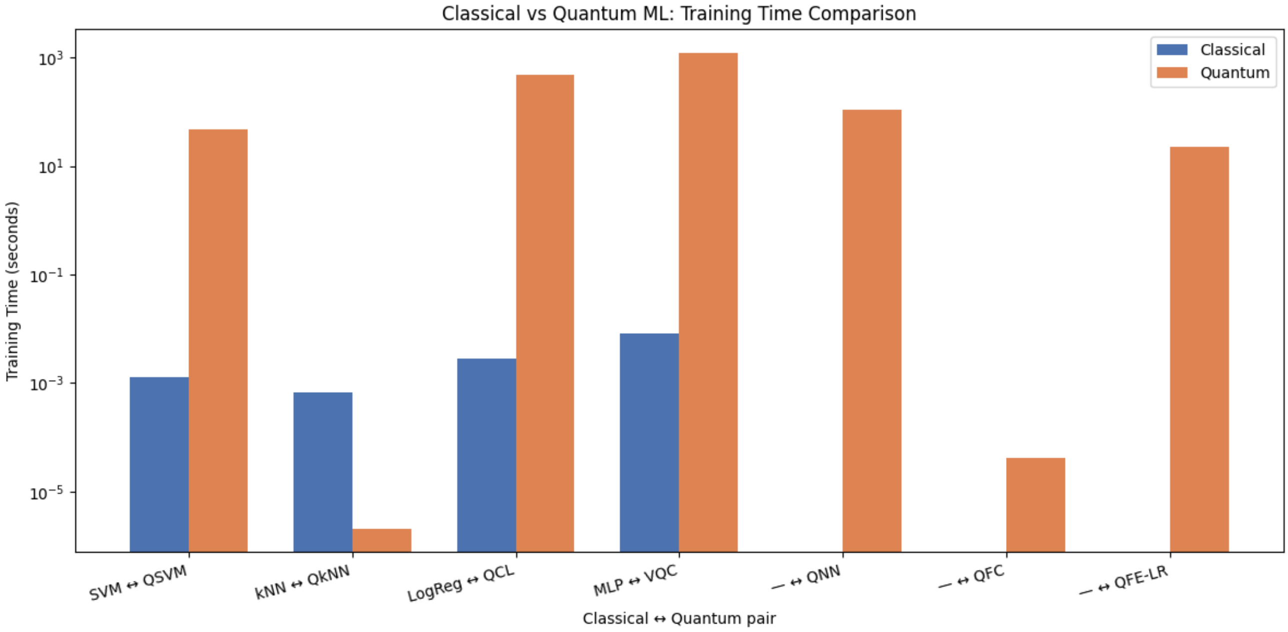

Fig. 1: Metrics and cost per model (n-var = non-variational; var = variational).

performance GPU and a low-latency link to the QPU would be expected to reduce the gap.

IV. Results and Discussion

The execution completed 33 of 33 jobs and 486 circuits with zero failures. The reported wall time of 2,227 s is dominated by queueing on shared cloud resources, a single gap of about 843 s separated two jobs and therefore does not represent the internal QPU execution time.

At aggregate level, classical models achieved a mean accuracy of 0.750 against 0.357 for the quantum models, and total classical training time was 0.013 s, roughly five orders of magnitude below the quantum 1.858 s.

Figure 1 shows a logarithmic scale graph was used due to the large variation in times between the classical computer, which had processing times in milliseconds, and the quantum computer, with up to 100 seconds in VQC (Variational Quantum Computing) on a normal scale. This would make the classical bars flatten and invisible next to the quantum ones.

Quantum times are so high for a few combined reasons. First, the quantum computer needs to repeat each circuit 1024 times (shots), since each measurement only gives one result per round. Second, the variational models (QCL, VQC, QNN) work in a loop going back and forth between IBM's quantum hardware in Washington and the classical computer that runs the optimisation, in this case, a Dell Precision notebook with an Intel i7 processor, 32 GB of RAM and 1 TB of SSD, running locally in Galway, Ireland, and each round trip adds to the network latency. Third, a large portion of the total recorded time is shared queueing in the IBM cloud, not actual quantum processing. In other words, the high time does not mean that the "quantum calculation" itself is slow, but rather that the combination of repetitions, machine-to-machine communication, and queueing makes the process, in practice, much more costly than its classical equivalent. It is important to remember that what counts in processing cost on IBM computers is the time spent running on the Heron r2 quantum processors.

In total, two runs were performed on the IBM quantum computer, but dozens of tests were conducted running the application with the AerSimulator until it was certain that the application was in real-world conditions to run on the quantum processor. The AerSimulator ran locally on the local computer without incident, only basic problems related to bugs that were corrected before running definitively on the quantum machines, where two runs were performed, leaving 3 minutes and 35 seconds of the available 10 minutes, which were ultimately left as a possible contingency.

A. Predictive Performance

In every classical×quantum pair the classical model equals or surpasses its quantum counterpart. Logistic Regression reaches perfect accuracy (1.000) on the test set while its quantum pair QCL scores 0.333; the only tie is k-NN↔QkNN (both 0.667), which requires a caveat. QkNN registered ran_on_backend = false: it fell back to the classical Gaussian-kernel proxy declared in the configuration and was not executed on the QPU, so it should be read as a classical reference. Among models that ran entirely on hardware, the best was VQC at 0.500.

The confusion analysis is consistent with the pre-processing: Logistic Regression produces a perfect diagonal, whereas QkNN identifies Setosa and Versicolor correctly but classifies every Virginica sample as Versicolor. This error follows directly from the PCA reduction from four to two dimensions, which brought the Versicolor and Virginica classes closer together. Across precision, recall and F1-macro, Logistic Regression dominates in all directions, while QNN and QFE-LR are the weakest (scores between 0.111 and 0.167) and the SVM/kNN/MLP trio is homogeneous at 0.667/0.556. Table IV consolidates the per-model metrics.

At class level the behaviour is consistent with the geometry of the reduced feature space: Setosa, which is linearly separable from the other two species, is recovered by almost every model, whereas the overlap introduced between Versicolor and Virginica by the two-component projection is where the weaker quantum models lose their accuracy. This

TABLE V. METRICS AND COST PER MODEL (N-VAR = NON-VARIATIONAL; VAR = VARIATIONAL).

Model	*Paradigm*	*Acc.*	*F1*	*Train (s)*
LogReg	Classic	1.000	1.000	0.003
SVM	Classic	0.667	0.556	0.001
kNN	Classic	0.667	0.556	0.001
MLP	Classic	0.667	0.556	0.008
QkNN	Q (n-var)	0.667	0.556	~0.000
VQC	Q (var)	0.500	0.389	1207.3
QSVM	Q (n-var)	0.333	0.333	46.7
QCL	Q (var)	0.333	0.278	473.7
QFC	Q (n-var)	0.333	0.333	~0.000
QNN	Q (var)	0.167	0.133	107.8
QFE-LR	Q (n-var)	0.167	0.111	22.9

indicates that the limiting factor is the information discarded in pre-processing rather than the quantum execution per se.

B. Computational Cost

Accuracy favours the classical models, which occupy the ideal corner of high accuracy at negligible time, while the quantum models running on hardware spread towards longer times and lower accuracy, with VQC isolated at about 1,207 s of training. On a logarithmic scale, classical models train in milliseconds and variational models in thousands of seconds; at inference, QFC remains expensive at 187 s because it recomputes fidelities in hardware for every prediction, and QCL reaches 91 s, against fractions of a second classically.

Expressed as a quantum/classical time ratio per pair, the overhead is extreme: 36,721× for SVM→QSVM, 146,942× for MLP→VQC and 168,759× for LogReg→QCL. The only pair where the quantum side appears faster, k-NN→QkNN, is again the classical proxy and does not represent real quantum execution.

Taken together, these measurements place every classical model in the ideal region of the accuracy-versus-time plane, high accuracy at negligible cost, while the hardware-executed quantum models scatter towards longer times and lower accuracy. The separation is not marginal: it spans four to five orders of magnitude in time without any compensating gain in predictive quality, which is the central empirical finding of the study for this class of small problem.

C. Variational Models

The low variational accuracy is explained by flat convergence curves rather than by any processing failure. For QCL and VQC the objective function varies between 11.3 and 12.0 over eight iterations without a consistent drop, and for QNN it remains at 1.25, consistent with the iteration ceiling imposed by the hardware guards (eight COBYLA rounds). Transpilation to QASM shows that only the variational models QCL, VQC and QNN reach depth 2; greater depth did not translate into greater accuracy, since QkNN at depth 0 was the most accurate (0.667) and QNN at depth 2 the least (0.167).

Separating the quantum models by type, the non-variational average total computation time (training + inference) is 83.9 s against 643 s for the variational group, making variational models about 7.7× slower on average. The non-variational QFC is an exception with a 187 s inference cost, showing that quantum cost is not reducible to the variational/non-variational dichotomy alone.

Each execution triggers a transpilation step that lowers the model to a hardware-native circuit (Qiskit to QASM); circuit depth measures how long that circuit is and therefore how many quantum gates it contains. In this experiment depth was not a predictor of accuracy: the depth-0 QkNN proxy was the most accurate while the depth-2 QNN was the least, so the additional circuit complexity of the variational models bought execution time rather than predictive quality under the eight-iteration ceiling.

TABLE IV. PAIRED COMPARISON OF ACCURACY AND TRAINING TIME.

Pair (classical→quantum)	*Cl. Acc.*	*Q. Acc.*	*Gap*	*Time ratio*
SVM → QSVM	0.667	0.333	0.333	36,721×
kNN → QkNN	0.667	0.667	0.000	0.003× (proxy)
LogReg → QCL	1.000	0.333	0.667	168,759×
MLP → VQC	0.667	0.500	0.167	146,942×

Table VI shows the available machines for running, all with the same capabilities. The selection was made thanks to the IBM_BACKEND_NAME=least_busy function, which chooses the least requested machine at runtime. The machine automatically selected was ibm_marrakesh.

TABLE VI. IBM QUANTUM COMPUTERS AVAILABLE FOR USE

Computer	*Quantum Processor Unit (QPU)*	*Qubits*
ibm_kingston	Heron r2	156
ibm_fez	Heron r2	156
ibm_kingston	Heron r2	156
ibm_marrakesh	Heron r2	156

D. Classical vs Quantum Paired Analysis

The largest accuracy gap is LogReg→QCL at +0.667 in favour of the classical model, followed by SVM→QSVM at +0.333 and MLP→VQC at +0.167; the k-NN→QkNN gap is zero but corresponds to the classical proxy. In summary, the classical paradigm wins three pairs and ties one, and the tie is not a genuine quantum execution. Table V details these pairs, where the time ratios reach factors of 10^4–10^5×.

V. CONCLUSION

Analysing the hypotheses formulated in the introduction, the null hypothesis (H_0) is rejected: the mean accuracy difference of 0.750 versus 0.357 and the time difference of up to about 169,000× are significant, and the alternative hypothesis (H_1) is corroborated. The observed quantum disadvantage, however, stems from the combination of NISQ noise, the eight-iteration ceiling for variational models, and the reduction of the problem to two dimensions, not from any inherent impossibility of quantum methods. The results should therefore be read as a controlled case study rather than as proof of the general superiority of any paradigm.

The main limitations are the small test set (six samples, so each hit or miss is worth about 0.167 accuracy and statistical power is low); the PCA reduction to two components, which discards information and approximates Versicolor and Virginica; the hardware-guard ceiling that prevented variational convergence; wall times that include queueing and network latency rather than pure QPU time; and the QkNN proxy, whose best quantum result was not run in hardware. Within these bounds, classical models showed superior accuracy at an extreme quantum computational cost of 10^4–$10^5\times$, without a proportional return in predictive quality.

On the methodology used, the quantum advantage did not materialise on a small problem such as Iris, and at a cost on the order of US$100 per minute of QPU time such use is impractical for this class of problem. Variational quantum circuits are currently the state of the art for error mitigation in the NISQ era, but their hybrid architecture, combined with the limited numerical power of the local notebook and the network latency between Ireland and Washington, produced a large performance discrepancy. Future researchers adopting this architecture should pre-evaluate the classical machine used for integration and, ideally, employ high-performance GPUs or servers for the classical part of the loop.

Two design choices dominate the outcome: the use of only two qubits, imposed by the one-qubit-per-variable mapping, which severely under-uses the main computational resource of these machines, and the number of shots, where a classical computer needs a single pass while a quantum computer requires many rounds of measurement to obtain an accurate result. Increasing the number of qubits and shots would raise both the expressivity and the precision attainable, at additional cost. HSBC's announcement [33] of financial gains using quantum computers gave no implementation details, but it is safe to say that simple two-qubit models like those studied here were not the source of that advantage.

Finally, the use of PCA was a pragmatic choice to keep execution within the ten-minute QPU window rather than a methodological necessity. It would be valuable to repeat the experiment without PCA, using more qubits, to re-examine the QkNN confusion between Virginica and Versicolor and to assess whether reducing a small dataset such as Iris from four to two components makes a significant difference. The source code is available at [37].

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